



Comsats University Islamabad, Lahore Campus

Lab Assignment <3> – SPRING 2025

Course Title	Software Testing	Course Code	CSE455	Credit Hours	3(2,1)
Instructor/s	Ms. Yella Mehroze	Program Name	BSSE		
Semester	8 th	Batch	FA21	Section	A
				Date	March 28, 2025
Time Allowed	1 week	Maximum Marks			
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Question Statement

[CLO-2]; Bloom Taxonomy Level: <Applying>

Write code in Java having errors mentioned in the document "coding standards & guidelines" and try to resolve them.

In the below driver class, code contains:

1. **Data reference error:** at line # 4, *num* variable is declared but not initialized and on the next line, it is referred.
2. **Array Index Error:** at line # 8, accessing an index out of bounds (valid range: 0-4)
3. **Computation Error:** at line # 11, division by zero is performed. At line # 15, *double* and *int* are added and their conversion to *int* type is not allowed.
4. **Comparison Error:** at line # 13, assignment operator (=) is used instead of comparison operator (==).

```
1 // Filename: ErrorProneCode.java
2 public class ErrorProneCode {
3     public static void main(String[] args) {
4         int num; // Uninitialized variable usage
5         System.out.println("Number: " + num); // Error: num is not initialized
6
7         int arr[] = new int[5];
8         arr[5] = 10; // Error: Array index out of bounds (off-by-one error)
9
10        int x = 5, y = 0;
11        int result = x / y; // Error: Division by zero
12
13        boolean flag = (x = 10); // Error: Assignment instead of comparison
14
15        int sum = 5 + 2.5; // Error: Type mismatch (adding int and double)
16        System.out.println("Sum: " + sum);
17    }
18 }
```

Errors produced by compiler:

```
ErrorProneCode.java:13: error: incompatible types: int cannot be converted to boolean
    boolean flag = (x = 10); // Error: Assignment instead of comparison
                   ^
ErrorProneCode.java:15: error: incompatible types: possible lossy conversion from double to int
    int sum = 5 + 2.5; // Error: Type mismatch (adding int and double)
                   ^
2 errors
```

I fixed the issues by following the actions:

1. Uninitialized variable is initialized (num = 0)
2. Used a valid index arr[4] instead of arr[5]
3. Ensured y is not zero before performing division
4. Used == instead of = operator
5. Used *double* for mixed-type arithmetic

```
1 // Filename: FixedCode.java
2 public class FixedCode {
3     public static void main(String[] args) {
4         int num = 0; // Initialize variable before use
5         System.out.println("Number: " + num);
6
7         int arr[] = new int[5];
8         arr[4] = 10; // Corrected: Using valid array index
9
10        int x = 5, y = 1; // Avoid division by zero
11        int result = (y != 0) ? (x / y) : 0;
12        System.out.println("Result: " + result);
13
14        boolean flag = (x == 10); // Corrected: Used comparison instead of assignment
15
16        double sum = 5 + 2.5; // Corrected: Use double type for mixed operations
17        System.out.println("Sum: " + sum);
18    }
19 }
20
```

Output:

```
Number: 0
Result: 5
Sum: 7.5
```